

Use of ICT

Candidates may use a word processor to produce their report, although they will not gain any extra marks for doing so.

In order to ensure that candidates demonstrate their understanding of the principles and techniques involved in analysing data, the use of ICT, eg spreadsheets, may not be used for analysing data for this unit.

Draft work

Candidates should do a variety of practical work during the course so that they develop the necessary skills to succeed in this unit. They should not, therefore, submit draft work for checking. However, teachers should check candidates' plans for health and safety issues before they implement the plan.

Neither the plan nor any practical work submitted for this unit should be returned to candidates for them to improve it.

How Science Works

This unit will cover the following aspects of how science works as listed in *Appendix 6: How science works* 2, 3, 4, 5, 6, 8, and 9.

12.2 Assessment information

Introduction

Candidates must produce a written plan for an experiment. They must also produce a laboratory report for an experiment that they have carried out. The experiment that they carry out may be based on the plan that they have produced; alternatively, the experiment that they carry out may be based on a plan that is either centre-devised or Edexcel-devised.

Plan

Students should not be given advanced details of the plan that they will carry out; they will be expected to draw on their experience of practical work that they have completed during the course for this assessment. Students should not take into the classroom any materials for this assessment.

If more than one group of students take this assessment at different times, then the groups must submit different plans for assessment to prevent plagiarism.

Centre-devised plans and experiments will not required Edexcel's approval; however, centre devised assessments must ensure that students have the opportunity to gain all the marks in the mark scheme.

If teachers are going to mark the plan they should not provide students with feedback on their plan until they have carried out their experiment. At this stage teachers could either:

- i) photocopy the plan, mark the original plan and provide students with the photocopy in the laboratory so that they can carry out their plan
- ii) collect in the plan, not mark it and return it to students in the laboratory under supervised conditions so that they can carry out their plan
- iii) mark the plan and ask students to do an experiment based on a different plan.

If teachers are not going to mark the plan they should collect the plan and check its feasibility. At this stage the teacher could either:

- i) return it to students in the laboratory under supervised conditions so that they can carry out their plan
- ii) ask students to do an experiment based on a different plan.

Practical work	Students will not need to take any documents into the laboratory for the practical aspect of this assessment although they may bring a scientific calculator. Teachers should issue students with the (unmarked) plan of the practical that they are to carry out. If necessary, teachers may allow students to analyse results under supervision in the next lesson. In this situation, teachers must collect in the work produced by their students. Teachers should not mark the practical work. In the following lesson, the documents should be returned to students under supervised conditions. Students should not have access to any other sources of information while they are completing the analysis of their results. Teachers who opt for internal assessment should mark the practical work after students have completed the analysis of their results.
Assessing work	The assessment criteria are shown in the next section. Each criterion scores a maximum of 1 mark. The criteria are not hierarchical. The assessment of planning, recording and analysis is based on written evidence in the form of a report.
For centres marking the written report	The written evidence should be annotated. This can be done by writing the skill code eg A15 near to the appropriate section of the report and ticking the box A15 on the grid below. The marks given for the report should be submitted to Edexcel on the mark sheets as shown in the <i>Edexcel Coursework Guide</i> that will be available on the Edexcel website www.edexcel.org.uk.
For centres not marking the written report	The written report should be submitted to Edexcel as discussed in the <i>Edexcel Coursework Guide</i> that will be available on the Edexcel website www.edexcel.org.uk.
Supervision	Students must work on their own for each part of this assessment. All aspects of this assessment must be done under supervised conditions.

Assistance for students

Students may require assistance whereby the teacher needs to tell the student what they have to do. Assistance in this respect carries a penalty. The teacher should record details of any assistance provided on the student's work.

Important reminder

Students should submit their work for assessment **once only**. **Neither** the plan **nor** the experiment should be given back to students to be improved